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Electrical Engineering, Mathematics - III (MTH 24231)
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124 Scholar No 254 Questions (Attempt any Three):

- (1) For the function $f(x) = x^3 - 2x + 5$, use the Bisection method to find a root in the interval $[a, b]$, where $(a = -3)$ and $(b = -2)$. Perform 5 iterations and write down the approximate root at the 5th iteration.
- (2) Consider the function $f(x) = x^2 - 29$ where $c = 29$. Use the Newton-Raphson method starting from an initial guess $x_0 = 3$ to find the root. Perform 4 iterations and provide the approximate root after the 4th iteration.
- (3) Given the polynomial $P(x) = x^4 - 6x^3 + x^2 + 9x - 5$, apply two iterations of Graeffe's Root Squaring method and write down the resulting polynomial.
- (4) Consider the polynomial $P(x) = x^3 - 4x^2 + 29x + 534$. Use two iterations of Lin-Bairstow's method with initial guesses $r = -9$ and $s = 87$ to update the estimates for r and s .